SDSC Summer Institute 2026

Performance Tuning

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Performance tuning

In this session,
we explore options to get more performance out of your tools.

Said another way, we look for ways to either

- Get the same problem solved faster.
- Get more problems solved in the same amount of time.

Why?

Why do you care?

- Getting results faster is of course always welcome
  - But does is make a material difference in real life?

Why do you care?

- Getting results faster is of course always welcome
  - But does is make a material difference in real life?
- For example:
  A tool takes 600ms to complete, and you reduce that to 10ms
  - If it is rarely used more than a few times a day, nobody will care
  - If it is something that is executed every time a process starts,
    it will likely make a noticeable cumulative difference
- Similarly, reducing a function runtime from 600ns to 10ns
  - If it is invoked once per process, it is unlikely anyone will notice
  - If it is invoked billions of times in an inner loop, it will be a game-changer

Why do you care?

- Getting results faster is of course always welcome
  - But does is make a material difference in real life?
- For example:
  A tool takes **1 hour** to complete, and you reduce that to **1 minute**
  - Even if rarely used, it will likely make a big difference

- Similarly, reducing a function runtime from **1 minute** to **1 second**
  - Even if it is invoked once per process, it will likely make a difference

Cost/benefit analysis

Getting results faster rarely comes for free

- Understand the cost
  - Are you paying it once?
  - Or will you have to pay many times to maintain the speedup?
- Are you able to pay the cost?
  - It may be expressed in either money or needed effort (or both)
- Are you willing to pay the cost?
  - Is it a nice to have?
  - Or is it transformational?
    - Without the speedup, it may take years to get the result!
    - One hour per iteration is doable. But I get distracted. A 1s reply allows for exploration.

Can I just use different HW?

HW vs SW

- Making software faster is obviously ideal
  - If done right, it is a **one time cost**
  - But **human time is expensive**
    (Modern AI tools can help, but they are not a silver bullet)
  - And may not always be possible
- Using different hardware to get results faster usually easier
  - Could even be cheaper
    - Especially if it is a one-off endeavor
  - But it is **a recurring cost**
    (i.e. it will add up over long periods of time)
  - And may still require some software changes

Types of hardware resources

- Datacenter vs consumer hardware
  - A high-end node in a datacenter is likely 10x to 100x faster than a laptop
  - You may be able to use identical software
- Clusters vs single nodes
  - Clusters can scale to 1000x of nodes
  - Pleasantly parallel problems may get 1000x faster (without any changes)
  - But most other problems will not scale linearly,
    and will likely hit a wall at one point.
    Plus, some software changes may be needed.
- Exotic hardware may provide even bigger speedups
  - E.g. GPUs, FPGAs, dedicated accelerators
  - Likely require major changes to the software

Can my SW be made faster?

Can SW be made faster?

- As we said, improving SW speed ideal
  - But not all SW can be made faster (on identical HW)

- First, you must **understand where your SW spends its time**
  - Both logical analysis and actual measurement recommended
  - A measurement-based approach is usually called **profiling**
- If 90% of the time is spent in a very small portion of the code, you better speed up that part!
  - This would be called **a hotspot**
- If compute is uniformly spread, it is very hard to get any improvement

Measure the right thing

- Use a realistic use case
  - Toy problems may not be representative
- Use the fully optimized executable
  - Compilers only do the right thing there (performance-wise)
  - Never benchmark a debug build
- Use realistic hardware setup
  - Using a single CPU core not the same as using the whole CPU (all cores)
  - A laptop CPU may behave differently than a datacenter one

Profiling modes

- Four major approaches
  - Timers in user code
  - Counting profilers
  - Sampling profilers
  - Hardware counters

Profiling modes

- Four major approaches
  - **Timers in user code**
  - Counting profilers
  - Sampling profilers
  - Hardware counters

The code owner is in the best position to understand the logical phases of an application.

Having a **coarse-grained idea of where** the applications spends its time is a great start!
And easy to measure how it changes with time and problem, as it is always on.

But usually needs to be paired with other methods.

Profiling modes

- Four major approaches
  - Timers in user code
  - **Counting profilers**
  - Sampling profilers
  - Hardware counters

Compilers can instrument the application, and **automatically insert** invocation counters and timing counters into your executable.

Gives you the **complete description** of the application's internal workflow, and how execution time is spent.

Unfortunately, this tends to add a non-negligible overhead, both slowing down the execution and **biasing the results**.

Profiling modes

- Four m
  - Time
  - **Cou**
  - Sam
  - Hard

```
   Incl | Incl Time |      Loop Exec |    Loop | Calltree=/[.]LOOP[.]
  Time% |           |                |   Trips |
        |           |                |     Avg |

 100.0% | 22,513.44 |             -- |      -- | Total
|-----------------------------------------------------------------------
1  99.9% | 22,481.98 |              1 | 7,001.0 | smain_.LOOP.8.li.335
||----------------------------------------------------------------------
2|  96.9% | 21,809.35 |          7,000 | 7,855.0 | stiff_ulgr_.LOOP.59.li.1905
|||---------------------------------------------------------------------
3||  89.9% | 20,231.88 |     54,985,000 |   134.3 | ulagra_.LOOP.1.li.81
||||--------------------------------------------------------------------
4|||   4.4% |    981.01 | 14,770,952,000 |     3.0 | mulqab_.LOOP.1.li.140
|||||-------------------------------------------------------------------
5||||   1.9% |    420.54 | 44,312,856,000 |     3.0 | mulqab_.LOOP.2.li.142
||||====================================================================
3||   1.0% |    214.79 |    423,696,000 |     3.0 | mulpru_.LOOP.1.li.85
||||--------------------------------------------------------------------
4|||   0.4% |     84.43 |  1,271,088,000 |     3.0 | mulpru_.LOOP.2.li.86
|||||-------------------------------------------------------------------
5||||   0.2% |     36.16 |  3,813,264,000 |     3.0 | mulpru_.LOOP.3.li.88
||||====================================================================
3||   0.7% |    155.40 |     54,985,000 |   134.3 | stiff_ulgr_.LOOP.60.li.1943
3||   0.6% |    142.89 |     54,985,000 |   134.3 | stiff_ulgr_.LOOP.83.li.2575
```

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Profiling modes

- Four major approaches
  - Timers in user code
  - Counting profilers
  - **Sampling profilers**
  - Hardware counters

You basically take a snapshot of where the code is many times during the execution of an application.

Taking a snapshot is extremely cheap, so it does not affect the performance of your application, resulting in an **unbiased measurement**.

It is a **great way to find hotspots**.
But does not help you identify the workflow that led there.

Profiling modes

- Four major ap
  - Timers in us
  - Counting pr
  - **Sampling p**
  - Hardware c

```
  Samp% |      Samp | Group
        |           |  Function

 100.0% | 263,922.0 | Total
|-------------------------------------
|  76.8% | 202,753.0 | USER
||------------------------------------
||  48.6% | 128,384.0 | mulqab_
||  14.6% |  38,557.0 | stiff_ulgr_
||   4.4% |  11,500.0 | mulpru_
||   3.3% |   8,671.0 | ulagra_
||   1.2% |   3,212.0 | msurfpair_
||   1.2% |   3,046.0 | interpolate_all_
||====================================
|  11.1% |  29,199.0 | HEAP
||------------------------------------
||  10.7% |  28,357.0 | calloc
||====================================
|   5.2% |  13,839.0 | BLAS
||------------------------------------
||   4.4% |  11,722.0 | _dgemm_
||====================================
|   2.7% |   7,220.0 | ETC: libf.so.2.0
||------------------------------------
```

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ring the execution of an application.

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lp you identify the workflow

Profiling modes

- Four major approaches
  - Timers in user code
  - Counting profilers
  - Sampling profilers
  - **Hardware counters**

Real hardware is composed of many different parts. It is often necessary to understand which part is your application stressing the most.

Most systems provide access to hardware-maintained counters. They may not be specific to your application, but are still representative if you are the only one stressing the system.

Exercise time

- Log into Expanse
- Checkout the git repository, if you did not do it already
- Go into 5.2_performance_tuning/exercises
- Do the exercise:
  - 1_profiling

Is there a better way to solve it?

You found a hotspot, now what?

- The best performance improvements usually come from better algorithms
  - I.e., Can you find a better way to solve the same problem?
- No generic advice I can give you there!
  - Unless the original implementation was a Proof of Concept only, it may need **serious research effort**!
  - But do **check if anyone else is doing the same thing**; they may have a better implementation than you do
  - Standard (or vendor-provided) libraries are definitely worth exploring

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  - But do check if anyone else is doing the same thing; they may have a better implementation than you do
  - Standard (or vendor-provided) libraries are definitely worth exploring
- We will assume you are stuck with the existing conceptual algorithm in the rest of these slides

Understand your HW

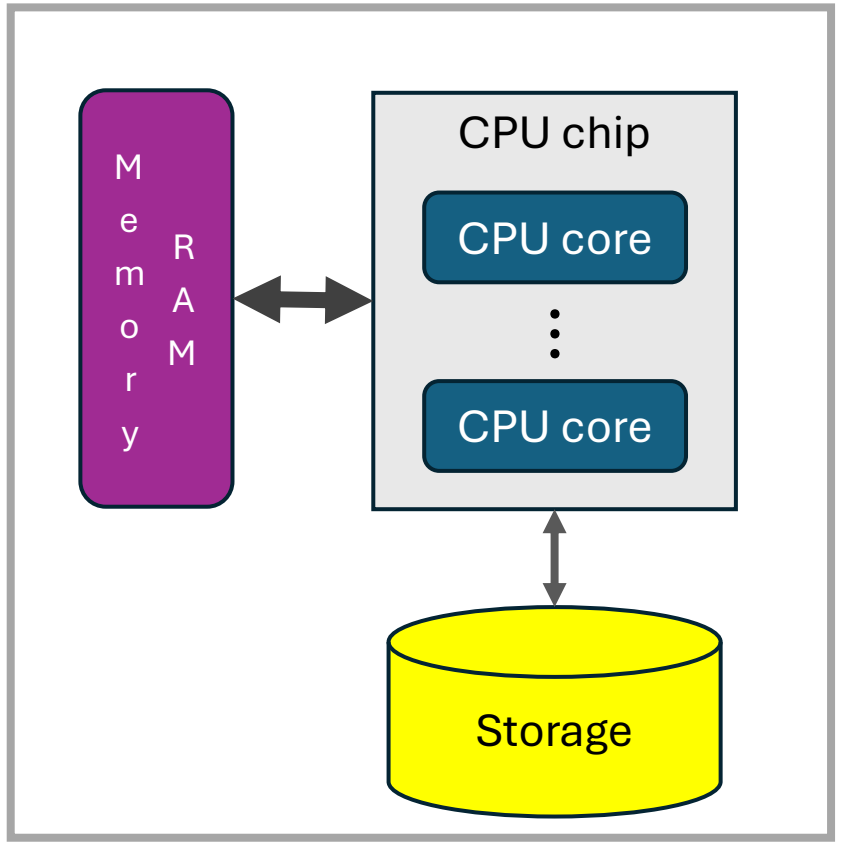
Software just drives the hardware

- The only way to make software faster (without changing the algorithm) is to **use the hardware better**
- But first you must understand
  - What hardware are you using?
  - How that hardware works?
- Spoiler:
  It is not a single, simple item

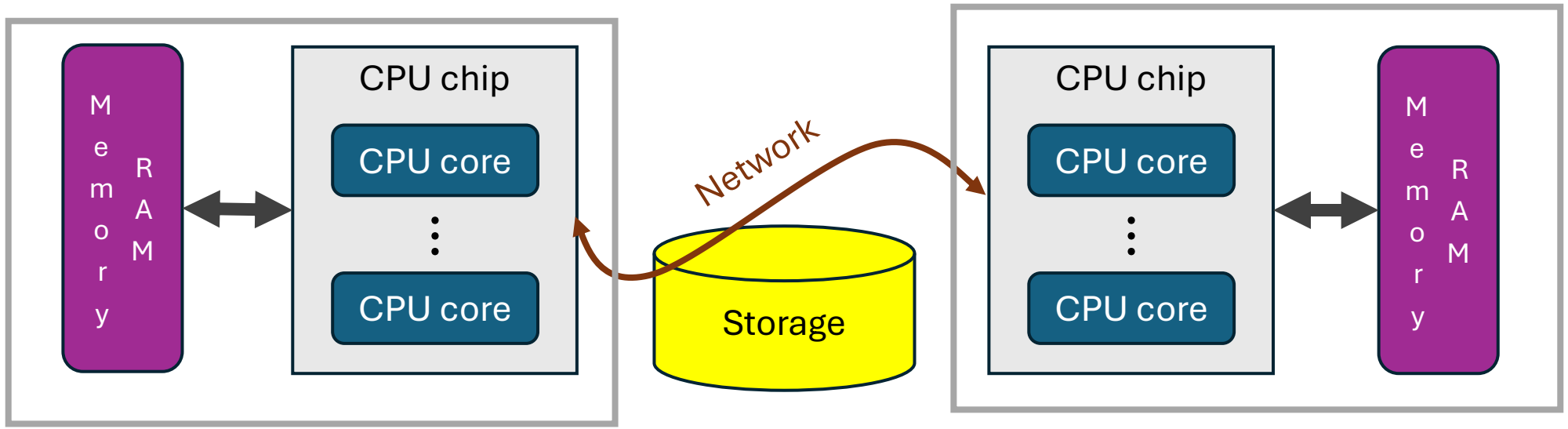
We will focus on CPU-only systems

- Luckily, virtually all CPUs are the same
  - At least conceptually
  - We will ignore some “weirdnesses” inherent to mobile systems
- GPU systems require a slightly different mindset
  - More on GPUs in another session
- FPGAs and dedicated accelerators are a completely different world
  - Many CPU concepts don’t apply there at all!

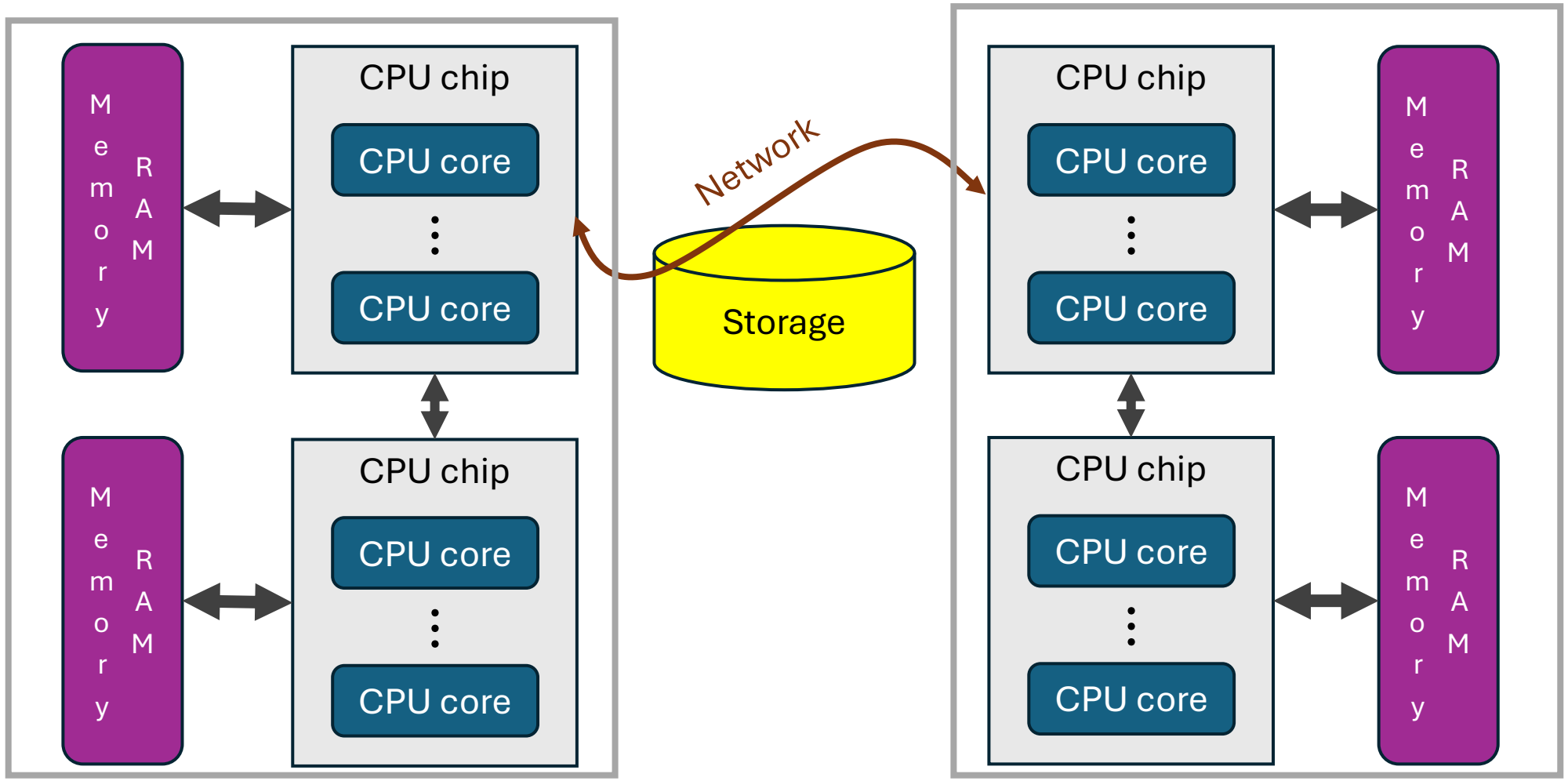
Very high-level view



Very high-level view - HPC



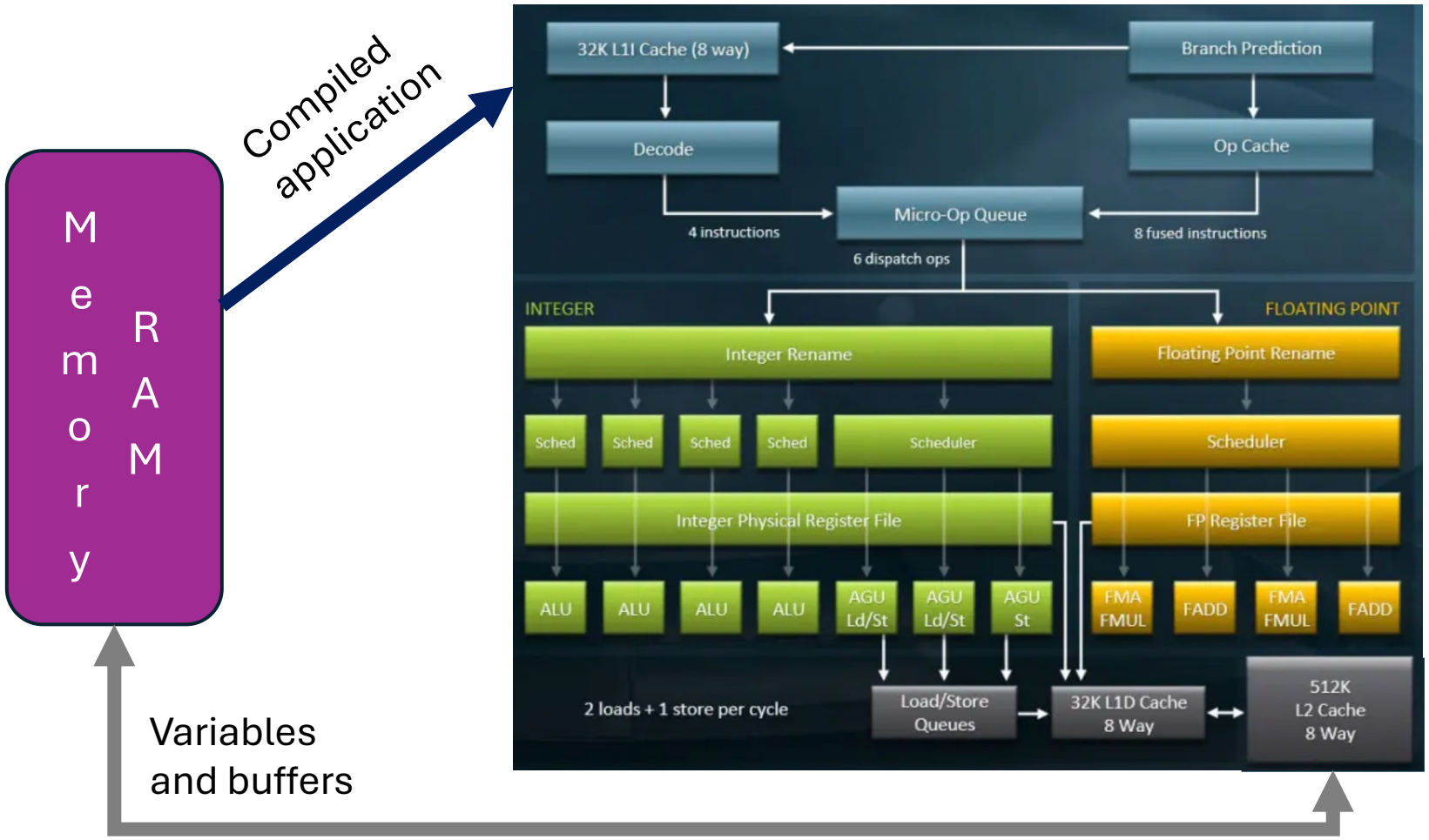
Very high-level view - HPC



Initial insight

- You do not use all the compute capability
  if you do not use all the CPU cores (e.g. OpenMP)
- When using multiple CPU cores,
  those cores compete for memory access
- Not all memory locations are the same
  NUMA = Non-Uniform Memory Access
  - Physically attached is faster
  - Accessing a memory location on another CPU much slower
  - Accessing memory on another node may need
    explicit network network communication (MPI)

In deep detail - Anatomy of a CPU core



The AMD “Rome” example

https://www.nextplatform.com/compute/2019/08/15/a-deep-dive-into-amds-rome-epyc-architecture/1659190

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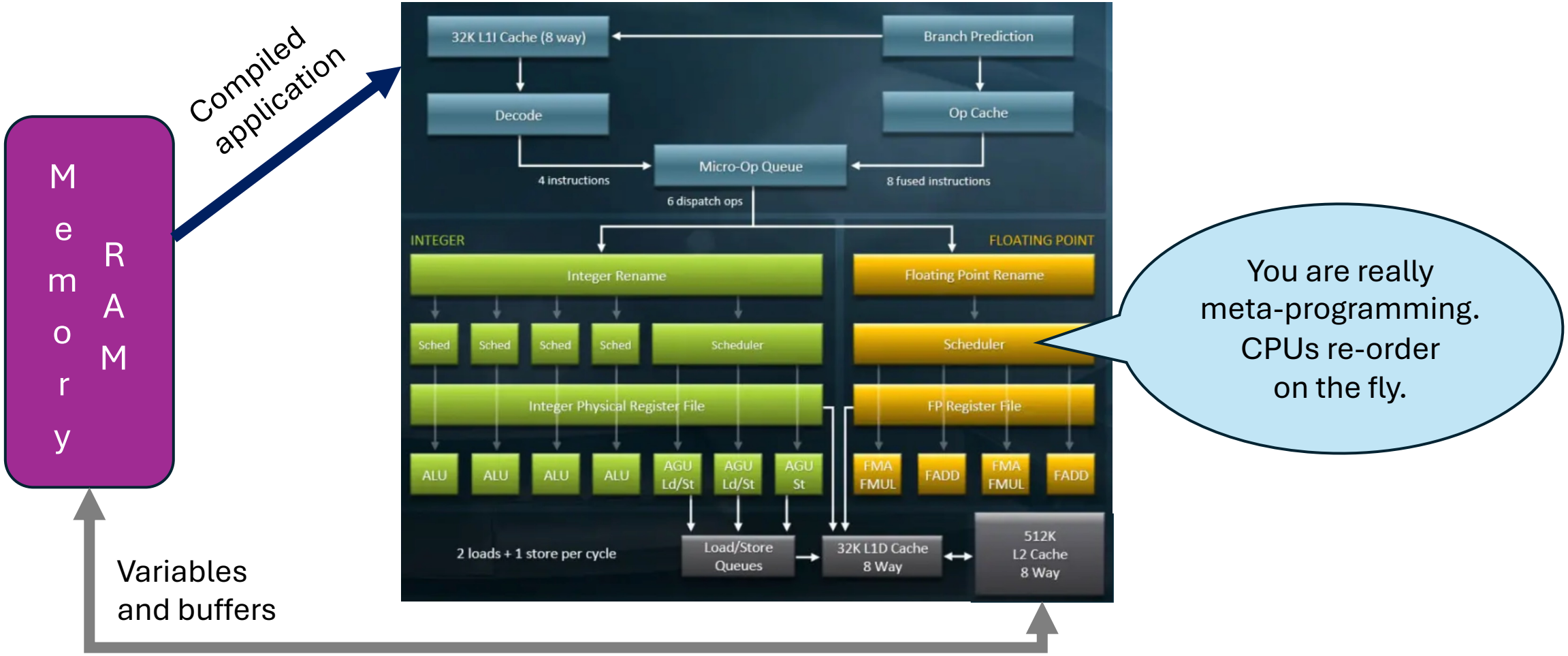

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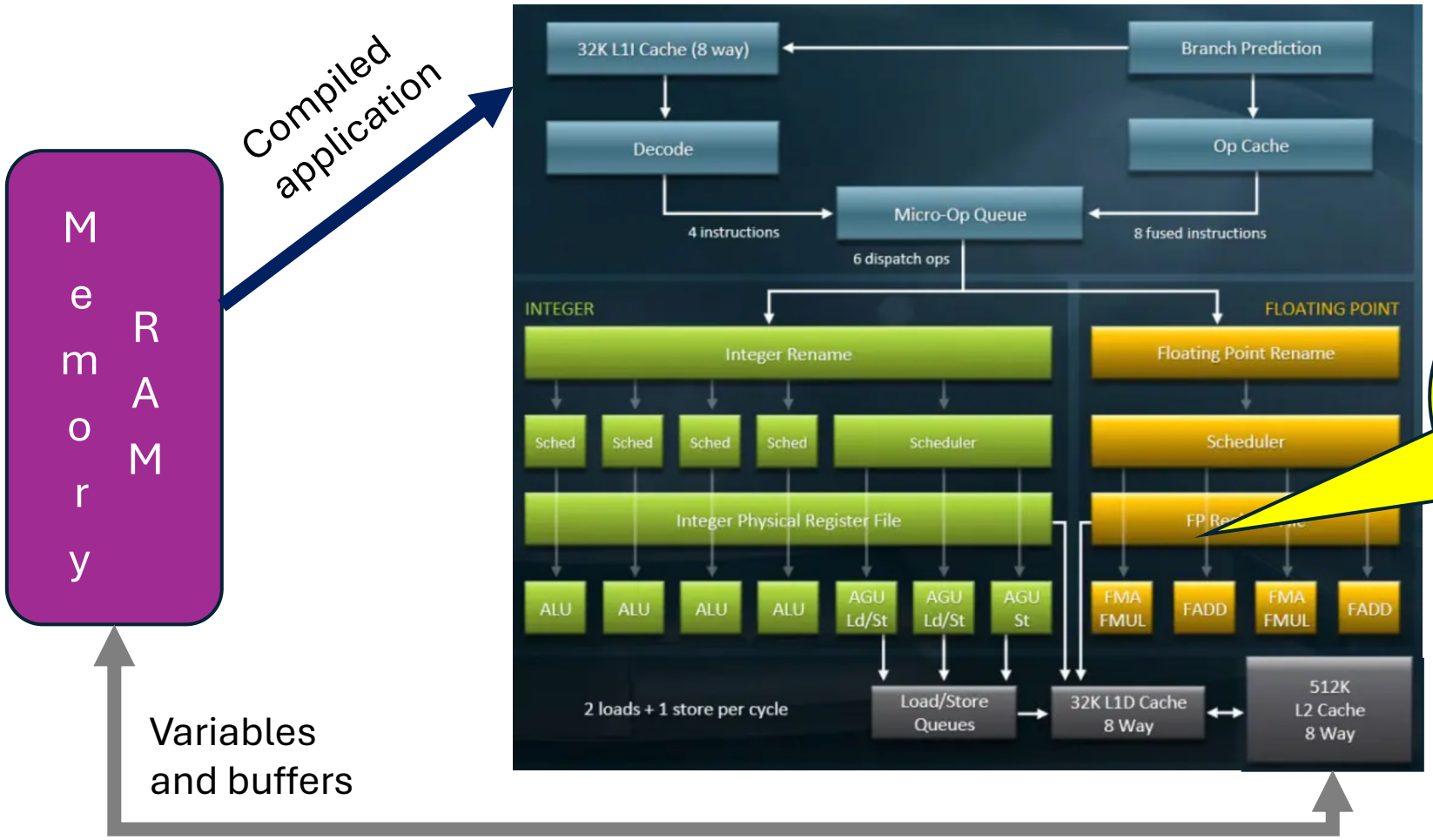

Need many independent compute steps to use all the compute units

Great:
a=sqrt(x);
b+=y*z;

Bottleneck:
a=sqrt(x);
b+=y*a;

https://www.nextplatform.com/compute/2019/08/15/a-deep-dive-into-amds-rome-epyc-architecture/1659190

In deep detail – Anatomy of a CPU core

Instructions are pipelined;
For most instructions, there is a big difference between how soon the result will be available (Latency) vs how often a new instruction can be scheduled (1/CPI):

Sqrt

Latency and Throughput

Architecture	Latency	Throughput (CPI)
Alderlake	12	6

FMA

Latency and Throughput

Architecture	Latency	Throughput (CPI)
Alderlake	4	0.5

https://www.intel.com/content/www/us/en/docs/intrinsics-guide/index.html

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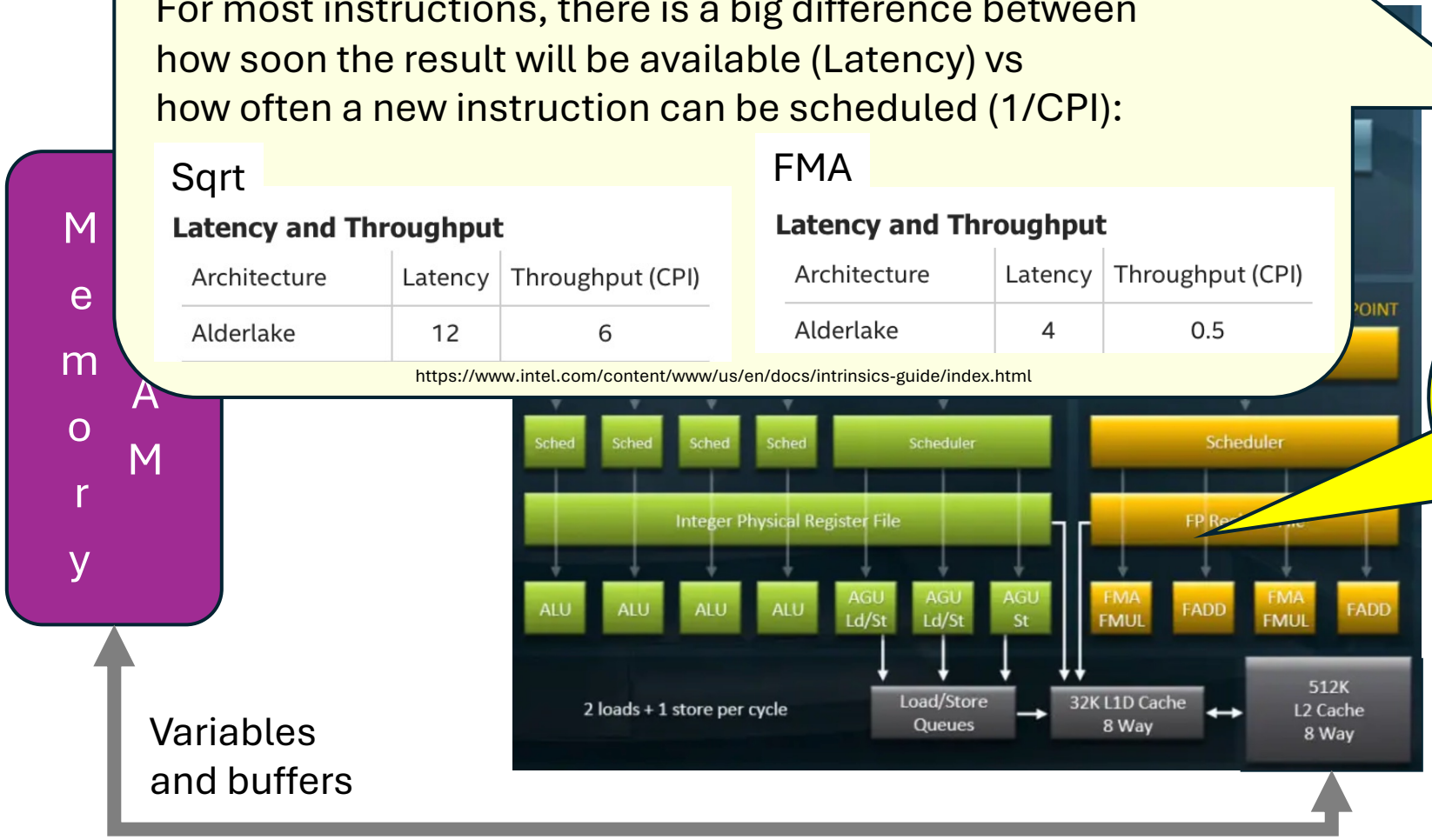

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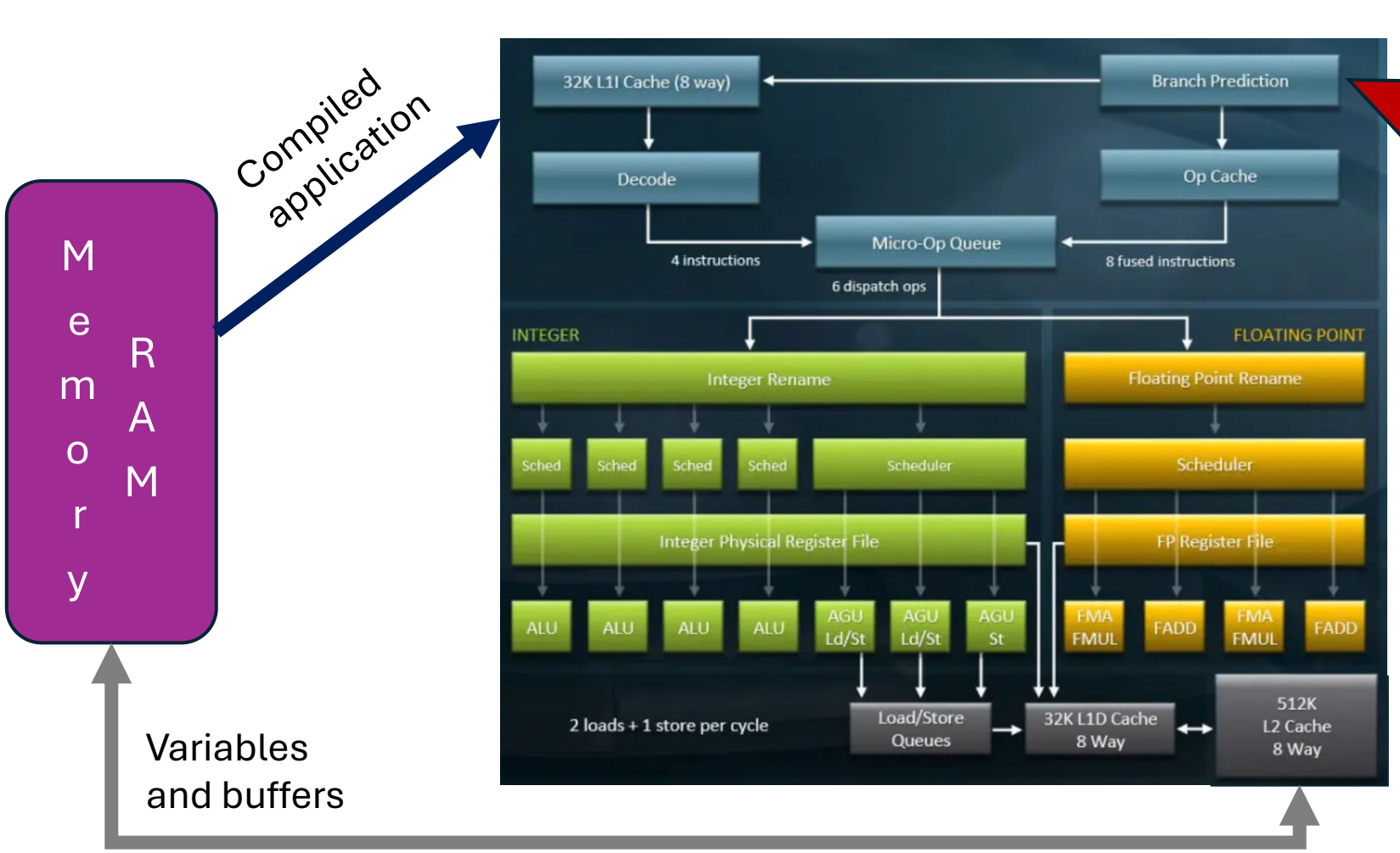
In deep detail - Anatomy of a CPU core

Good:
a=sqrt(x);
b+=1/sqrt(y);



Branching (**if** statements) breaks internal parallelism. **CPU will try to guess.**

Likely OK:
a=sqrt(x);
If (flag1) b+=1/sqrt(y);
else b+=sqrt(z);

Quite bad:
a=1/sqrt(x);
If (a<1) b+=1/sqrt(y);
else b+=sqrt(z);

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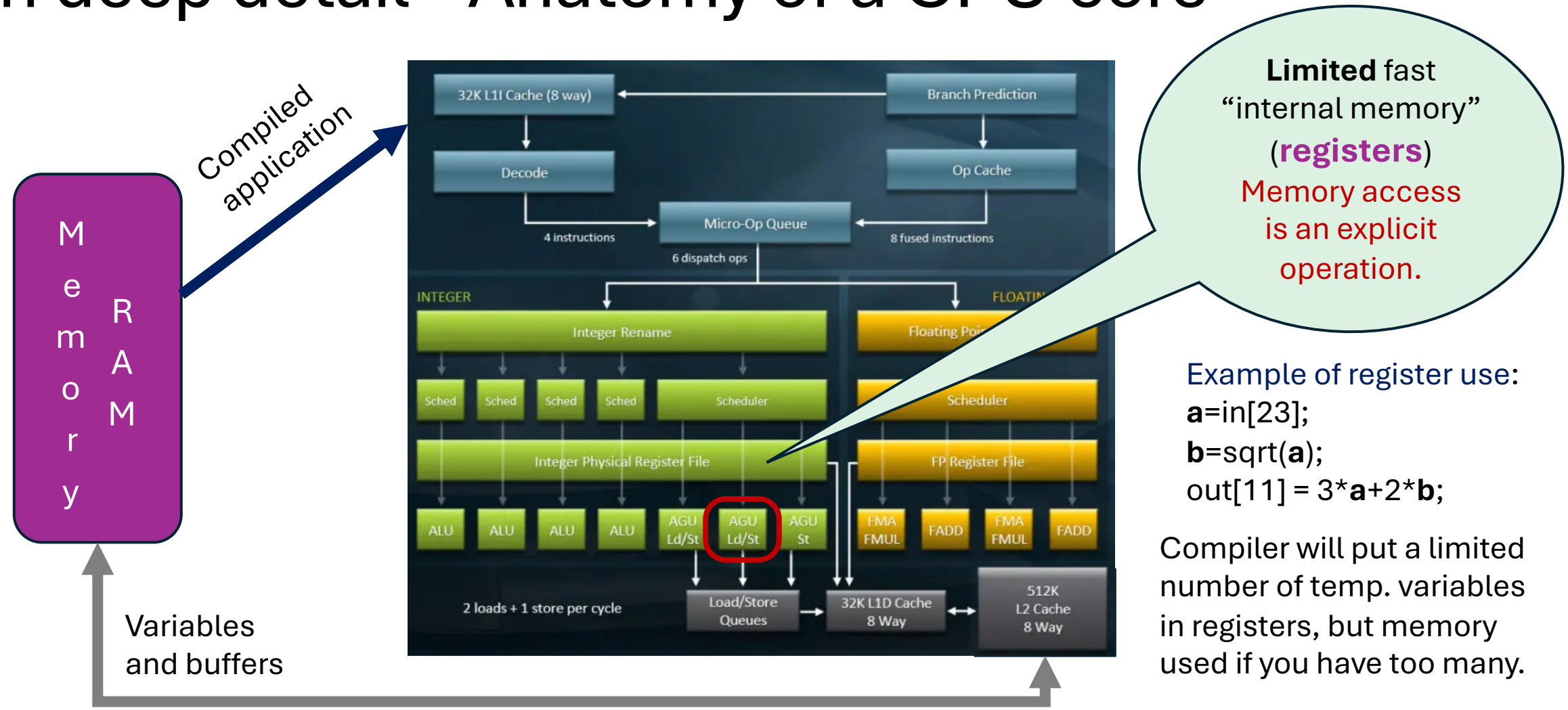
In deep detail - Anatomy of a CPU core



Example of register use:

a=in[23];
b=sqrt(**a**);
out[11] = 3***a**+2***b**;

Compiler will put a limited number of temp. variables in registers, but memory used if you have too many.

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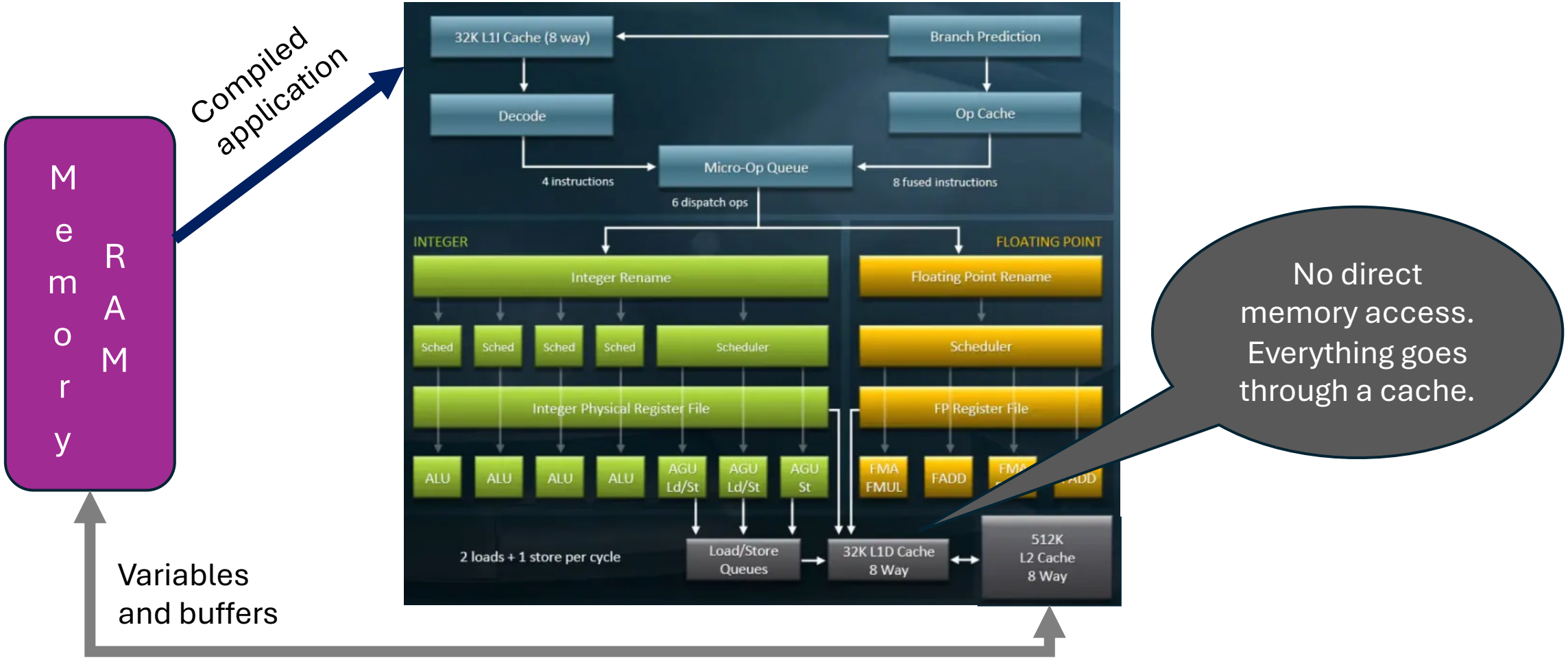

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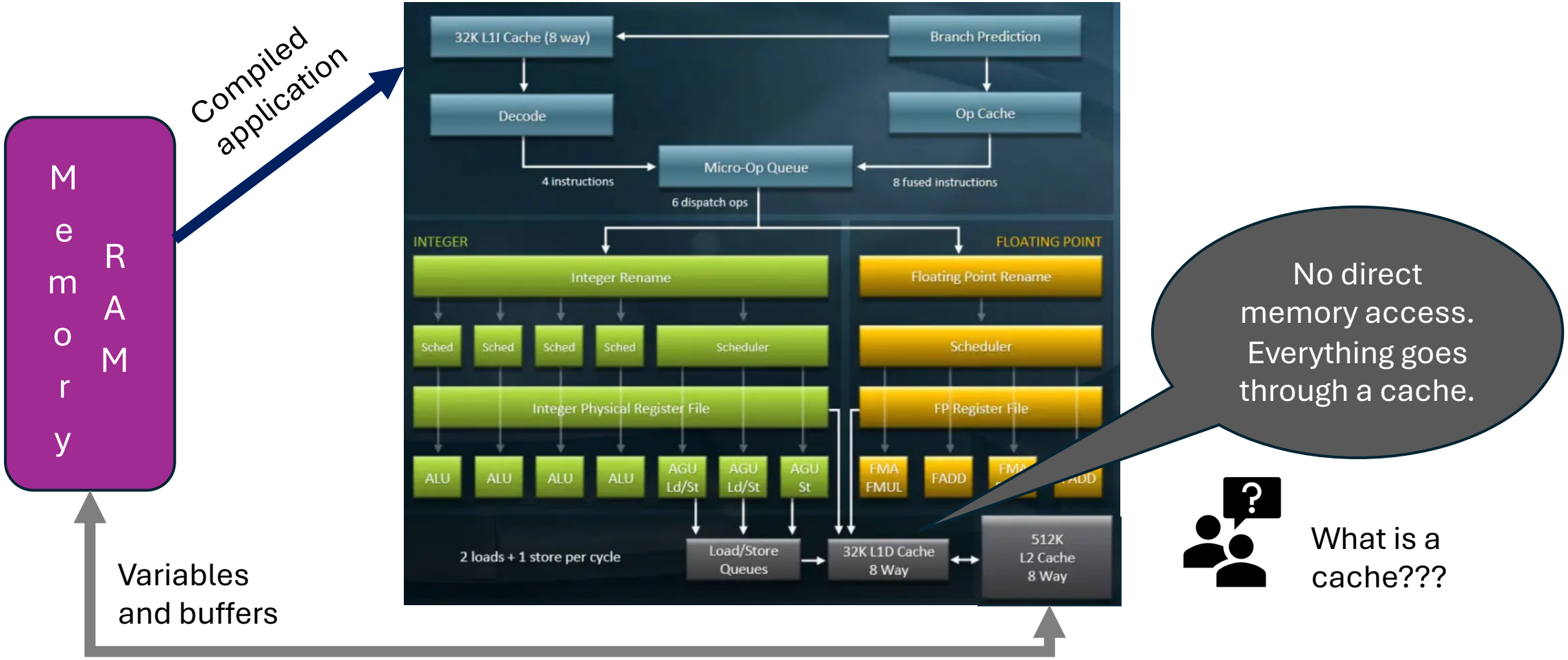


The cost of direct memory access

- Accessing information straight from memory is very expensive
  - You can do about **300 compute operations** in the time it takes to fetch a byte from main memory (RAM)!
- Classic examples, like AXPY, can only go as fast as memory can deliver
  - Compute is basically free there

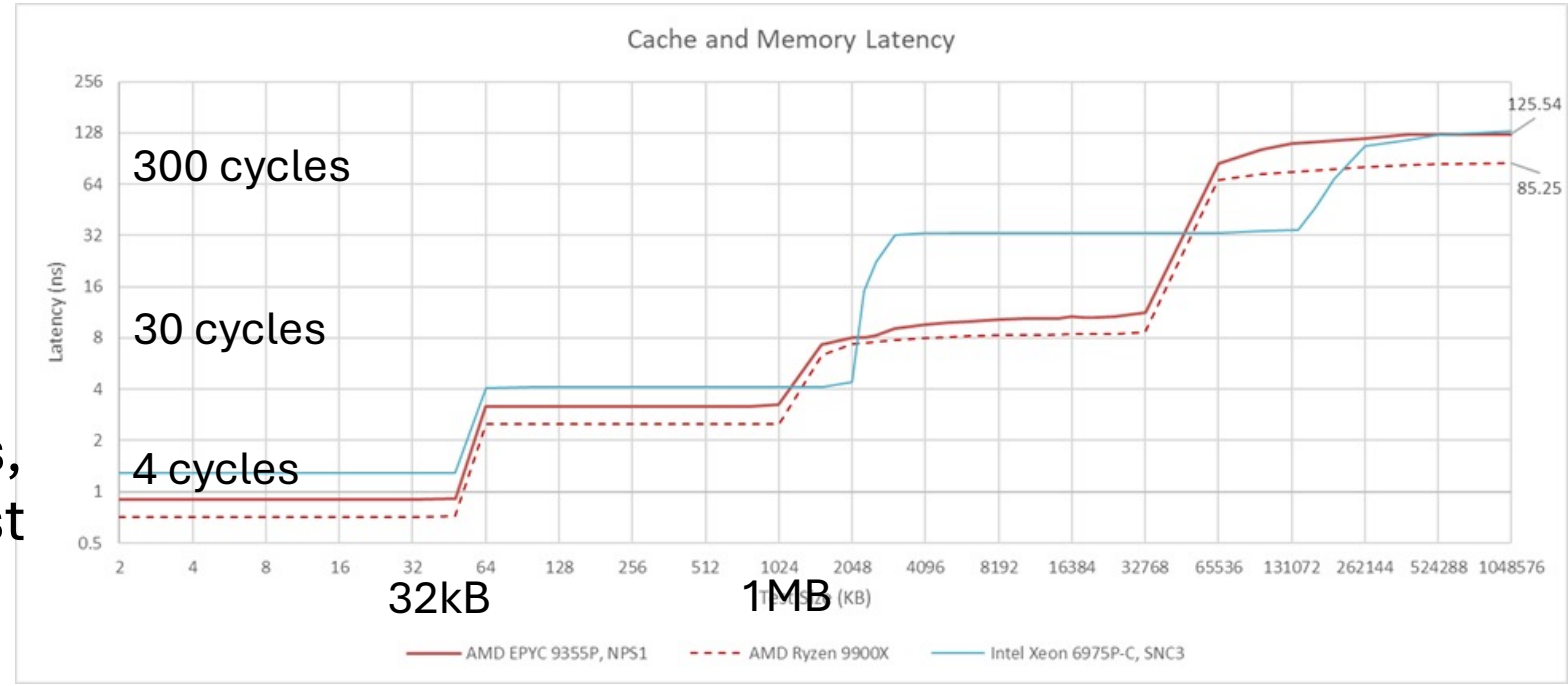
AXPY:
x[i] += a*Y[i]

Caches to the rescue

- Accessing information straight from memory is very expensive
  - You can do about **300 compute operations** in the time it takes to fetch a byte from main memory (RAM)!
- All modern CPUs thus contain some kind of **memory cache**
  - Access to data in the cache much faster
  - You typically waste **only 4-40 compute operations** (still not free, though)
  - Contains **most recently used data**
    No need to explicitly populate it,
    and data automatically purged to make space

Caches to the rescue

- Accessing information straight from memory is very expensive
- All modern CPUs thus contain some kind of **memory cache**
  - Access to data in the cache much faster
- Cache sizes are small (measured in **kBytes**)
  - Typically several levels, fastest level is smallest



The importance of memory reuse

- The only way to fully utilize the compute capabilities of modern CPU cores, is to **reuse the same data over and over again**
  - Ideally from registers
  - But L1 cache can be good enough if you can use internal parallelism
  - Everything else is memory-bound!
- Basic advice:
  - **Avoid** accessing the same buffer multiple times!
  - Try to extract all the information in one pass

Bad:

```
for i: x[i] = y[i]+z[i]
for i: x[i] += sin(y[i])
for i: x[i] = min(x[i],sqrt(z[i])
```

Great:

```
for i:
  a=y[i]; b=z[i]
  x[i]=min(a+b+sin(a),sqrt(b))
```

The importance of memory reuse

- The only way to fully utilize the compute capabilities of modern CPU cores, is to **reuse the same data over and over again**
  - Ideally from registers
  - But L1 cache can be good enough if you can use internal parallelism
  - Everything else is memory-bound!
- Basic advice:
  - Avoid accessing the same buffer multiple times!
  - If you need a **work buffer, keep it small**.
    - So it fits in L1 cache (**< 24kB** on most systems)
    - May not always be possible, but think hard if there is a way (Can you use a smaller el. type? fp32 vs fp64? bool vs int?)

Good:

```
for (s,e) in split_range(N,1k):
  x[s:e] = y[s:e]+z[s:e]
  x[s:e] += sin(y[s:e])
  x[s:e] = min(x[s:e],sqrt(z[s:e])
```

Understand the cache line

- Cache granularity is **not** at the byte level
  - A **cache line** is typically 64 bytes
  - The whole cache line is present in a cache, or none of its bytes are!
- **Strive for contiguous memory access**
  - Else, your effective work area explodes!

Example where “small index loop” does not fit in L1 cache.

```
struct MyEl {
  double d[7]; // 8*7 bytes
  bool f[7]; // 7 bytes
}; // 64 bytes total (1 wasted)

double sum(int i, int j, MyEl  b[1024]) {
  for el in b: el.d[0] = 0;
  for el in b: if (el.f[i]) el.d[0] += el.d[i];
  for el in b: if (el.f[j]) el.d[0] += el.d[j];
  double sum = 0;
  for el in b: sum += el.d[0];
  return sum;
};
```

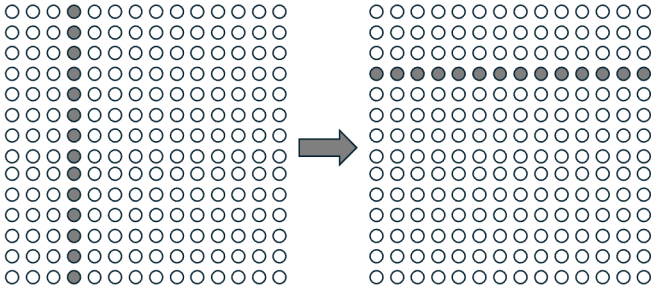
Understand the cache line

- Cache granularity is **not** at the byte level
  - A **cache line** is typically 64 bytes
  - The whole cache line is present in a cache, or none of its bytes are!
- Strive for contiguous memory access
- When writing, be careful of **false sharing**
  - If two cores access the **same** cache line, only one can own it
  - A write will always invalidate cache lines in other CPU cores (ping-pong effect)

Example of false sharing:

```
int cnt[...];
#pragma omp parallel
for (int i=0; i<N; i++) {
  if (sqrt(x[i]<1) cnt[omp_get_thread_num()]++;
}
```

Useful technique – 2D tiling



- Classic transpose example

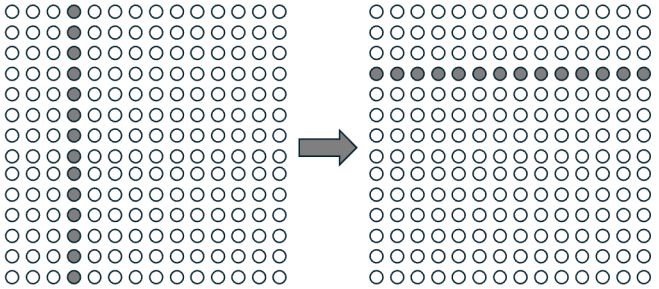
Naïve:

```
for (int i = 0; i < N; i++)
 for (int j = 0; j < M; j++) {
   B[i * M + j] = A[j * N + i];
 }
```

Reads a whole column.

We populate 64*M bytes in L1 cache.
If M is small, the next 7 iterations
will reuse the data from cache.
(Assuming double elements, i.e. 8 bytes)

Useful technique – 2D tiling



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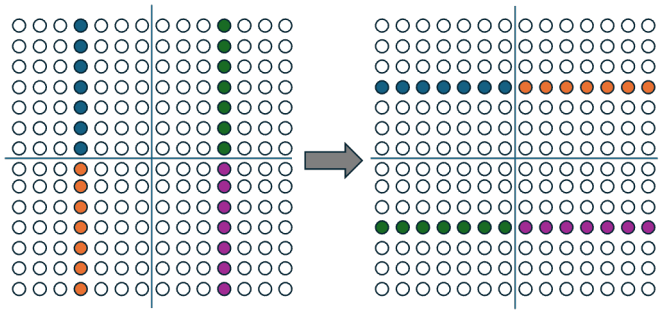
Reads a whole column.

We populate 64*M bytes in L1 cache.
If M is large (>500), we are reading those same 64*M bytes from slower memory again!

Useful technique – 2D tiling



- Classic transpose example

Transpose one small sub-matrix at a time.

Guaranteed to read from the same cache line.

Reads only T elements.

Naïve:

```
for (int i = 0; i < N; i++)
 for (int j = 0; j < M; j++) {
  B[i * M + j] = A[j * N + i];
 }
```

Tiled: (T decided at compile time)

```
for (int sj = 0; sj < N; sj += T)
 for (int sj = 0; sj < M; sj += T) {
  for (int i = si; i < si + T; i++)
   for (int j = sj; j < sj + T; j++) {
    B[i * M + j] = A[j * N + i];
   }
 }
```

(Assuming N and M divisible by T)

We access 64*T bytes in the first pass. By **picking** a small T, the next 7 passes will already have the data in L1 cache.
(Assuming double elements, i.e. 8 bytes)

Useful technique – 2D tiling

- Classic transpose example

Naïve:

```
for (int i = 0; i < N; i++)
 for (int j = 0; j < M; j++) {
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 }
```

Tiled:

```
for (int sj = 0; sj < N; sj += T)
 for (int sj = 0; sj < M; sj += T) {
   for (int i = si; i < si + T; i++)
     for (int j = sj; j < sj + T; j++) {
        B[i * M + j] = A[j * N + i];
     }
 }
```

Using OpenMP:

```
#pragma omp tile sizes(T,T)
for (int i = 0; i < N; i++)
 for (int j = 0; j < M; j++) {
  B[i * M + j] = A[j * N + i];
 }
```

Same semantics.
Tiling happens under the hood.
(Requires OpenMP 5.1+ compiler)

The first access problem

- Caches are great for speeding up (small) temp buffers
  - I.e., Reading back recently written data
- But most data not already in cache
  - Caches can still help!
  - Most CPUs use **prefetching**
    I.e. the CPU will guess what the next needed cache line will be and fetches it into Cache before you ask (in background)
  - If compute/memory ratio large, may make memory access free
- Prefetching works great for sequential access
  - E.g. AXPY-like

The first access problem

- Caches are great for speeding up (small) temp buffers
  - I.e., Reading back recently written data
- But most data not already in cache
  - Caches can still help!
  - Most CPUs use **prefetching**
- Prefetching works great for sequential access
- Does not work for pseudo-random access
  - For example, tree traversal
  - You can help by using explicit prefetch commands
    In gcc, it is called **__builtin_prefetch**

Sharing data between CPU cores

- Most modern CPU share a L3 cache among several cores
  - Much slower than L1 cache, but still significantly faster than main RAM
- When using parallel programming (e.g. OpenMP)
  - If your **read-only work area** is too big for L2, try at least to **fit it in L3 Cache**
  - Size of L3 cache often comparable to sum of L2 cache sizes, but you do not need to partition it
- Note: Not helpful for memory you write to

Modern AMD Epyc CPU

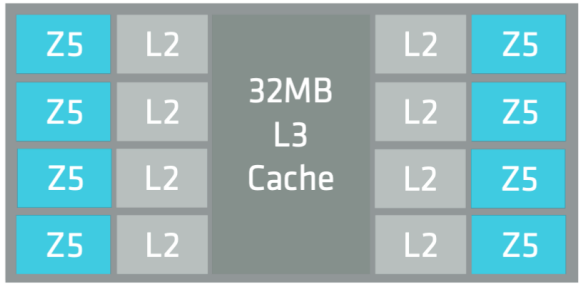

Figure 2: A 'Zen 5' CCD with 8 cores

https://docs.amd.com/v/u/en-US/5th-gen-amd-epyc-processor-architecture-white-paper

All modern CPUs do vector math

- Not emphasized in the previous diagrams is the fact that **all compute units are vectorized**!
- **Computing a 256-bit vector takes as much time as computing an 8-bit scalar**
  - Scalars are really just partially filled vectors, from a HW point of view
- Only way to make full use of CPU compute capabilities is write vector code

Scalar int32 mul		Vector int32x4 mul	
imul		**vpmuldq**	
Latency	CPI	Latency	CPI
3	0.33	3	0.5

On Zen 5, https://uops.info/table.html

How do we write vector code?

- For simple loops, the compiler takes care of everything
  - E.g. AXPY
- For more complex loop, compiler can still do the right thing, IF:
  - You **access consecutive memory locations**
  - Have no complex logic (if statements)

Likely vectorized:

```
for i:
  a=y[i]; b=z[i]
  x[i]=min(a+b+sin(a),sqrt(b))
```

Probably not vectorized:

```
for i:
  a=y[i*3]; b=z[i%7]
  if (w[i%25]>sin(b)) a+=b;
  x[i*3]=min(a+b+sin(a),sqrt(b))
```

https://gcc.gnu.org/onlinedocs/gcc-8.3.0/gcc/Vector-Extensions.html

How do we write vector code?

- For simple loops, the compiler takes care of everything
  - E.g. AXPY
- For more complex loop, compiler can still do the right thing, IF:
  - You **access consecutive memory locations**
  - Have no complex logic (if statements)
- For really complex logic, you may need to use explicit vector types
  - Not part of most language specs
  - But most modern compilers have native support for it

GCC example:

```
typedef int v4si __attribute__ ((vector_size (16)));

v4si a, b, c;

c = a + b;
```

https://gcc.gnu.org/onlinedocs/gcc-8.3.0/gcc/Vector-Extensions.html

How do I know if the code was vectorized?

- Most compilers have compile options to tell you if a loop was vectorized
  - Use **-fopt-info-vec** with gcc

```
$ gcc -O3 -fopt-info-vec -o axpy_o3 axpy.c -lm
axpy.c:19:5: optimized: loop vectorized using 16 byte vectors
axpy.c:93:5: optimized: loop vectorized using 16 byte vectors
axpy.c:19:5: optimized: loop vectorized using 16 byte vectors

$ gcc -O3 -fopt-info-vec -march=native -o axpy_o3 axpy.c -lm
axpy.c:19:5: optimized: loop vectorized using 32 byte vectors
axpy.c:93:5: optimized: loop vectorized using 32 byte vectors
axpy.c:19:5: optimized: loop vectorized using 32 byte vectors
```

- Too verbose to have it always on
  - But great for when developing now code or trying to optimize existing code
  - Worth checking across compilers and versions, too

How do I know if the code was vectorized?

- Most compilers have compile options to tell you if a loop was vectorized
  - Use **-fopt-info-vec** with gcc
- You can also look at the generated assembly
  - Use **-g -fverbose-asm -save-temps** with gcc
  - Look for **.s** files
  - Usually lots of helper code, too, so may not be trivial to find what you look for

```
$ gcc -O3 -g -fverbose-asm -save-temps -o axpy axpy.c –lm
$ more axpy.s
…
# axpy.c:20:       Z[i] = alpha * X[i] + Y[i];
     vmovups (%rdx,%rax), %ymm1
     vfmadd213ps    (%rcx,%rax), %ymm2, %ymm1
     vmovups %ymm1, (%rsi,%rax)
# axpy.c:19:    for
     addq    $32, %rax
     cmpq    %r8, %rax
     jne    .L5
```

Function calls and vectorization

- Complex/expensive functions should be vectorized on their own
  - Using any of the methods mentioned before
- But **cross-function vectorization may be more effective**
  - **Especially for simple functions**
    (but not only)

```
double rsqrtsum(double a, double b, double m) {
  return 1/sqrt(a)+m*1/sqrt(b);
}
...
for i:
  Z[i] = rsqrtsum(A[i], B[i], C[i]);
```

Function calls and vectorization

- Complex/expensive functions should be vectorized on their own
  - Using any of the methods mentioned before
- But **cross-function vectorization may be more effective**
  - **Especially for simple functions** (but not only)
- Compilers can only vectorize if they see the function content
  - A problem with separate compilation
  - Use **header-only inline** in C++, when possible

```
inline double rsqrtsum(double a, double b, double m) {
  return 1/sqrt(a)+m*1/sqrt(b);
}
...
for i:
  Z[i] = rsqrtsum(A[i], B[i], C[i]);
```

Function calls and vectorization

- Complex/expensive functions can be vectorized on their own
  - Using any of the methods mentioned before
- But **cross-function vectorization may be more effective**
  - **Especially for simple functions** (but not only)
- Compilers can only vectorize if they see the function content
- You can also use explicit vectorization at the interface

```
typedef double v4d __attribute__ ((vector_size (32)));
v4d rsqrtsum(v4d &a, v4d &b, v4d &m) {
  return 1/sqrt(a)+m*1/sqrt(b);
}
...
for i:
  Z[i] = rsqrtsum(A[i], B[i], C[i]);
```

Use all the CPU cores

- No single-core CPUs anymore
  - You must find a way to parallelize your problem
- You can do it in three orthogonal ways
  - Split in several independent subproblems (with minimal glue before and after)
    - You had an introduction to HTC a few days ago
  - Partition the problem in overlapping sub-problems
    - We discussed how to use MPI to deal with synchronization
  - Distribute the (sub-)problem over multiple cores
    - But remember NUMA and race conditions
    - We went over OpenMP a few days ago
- A combination of the 3 will give you the best results

Use all the CPU cores

- No single-core CPUs ap
  - You must
- You can
  - Spli
  - I
    -
  - D
    -
    - We wer
- A combination of the 3 wit e best results

But remember:
Main memory (and L3 cache) are shared
between CPU cores,
so the more CPU cores you use,
the slower access to memory will be.
(per core)

and after)

Exercise time

- Log into Expanse
- Checkout the git repository, if you did not do it already
- Go into 5.2_performance_tuning/exercises
- Do the exercises:
  - 2_memory
  - 3_vectors
  - 4_functions
- You can also try to optimize code from other sessions

In summary

- Make sure you understand where the application spends its time
  - Logical understanding of the code recommended
  - But actual measurement (profiling) required, too
- Make sure you use all the resources you have access to
  - All cores on the local CPU(s)
  - All the nodes you have access to
- You code is likely memory-bound
  - Look for register and Cache-friendly optimization options first
- Modern CPUs do vector math
  - Help the compiler generate full-width vector instructions
- A better algorithm may give you an even better speedup
  - See if anyone else is doing something similar (and consider libraries)